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Femtosecond laser photochemical reduction of silver ion solution for micro-welding of transparent and hard materials for solar cell packaging

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ABSTRACT

Transparent and hard materials are extensively employed in the fabrication of integrated devices. Ultrafast laser micro-welding technology serves as an efficient method for joining such materials. However, initiating multi-photon absorption in transparent materials necessitates high pulse energy, and a polished surface forming optical contact is crucial for ensuring reliable laser welding. Consequently, achieving high-quality connections of transparent materials under non-optical contact conditions presents a challenge. In this study, a silver ion solution was utilized as an intermediate layer for femtosecond laser micro-welding of glass and heterogeneous materials. The findings demonstrated that photochemically reduced silver nanoclusters increased the shear strength of glass to 27.36 MPa at a low input energy density (2.4J/cm^2). Furthermore, the quality of the connection and sealing reliability were assessed through thermal shock and waterproof evaluations. The results indicated that femtosecond laser welding assisted by a silver ion solution is an effective and energy-efficient technique for achieving high-quality connections of non-optically contacted transparent materials, offering a promising approach for applications such as solar cell packaging.

1. Introduction

Transparent and hard materials, such as glass and sapphire, possess exceptional optical properties, high mechanical strength, and remarkable corrosion resistance, making them widely applicable in the fabrication of electronic devices, biochips, and optical components[1–4]. In optoelectronic devices, establishing connections between diverse transparent materials is crucial for optical information transmission[5] and sealing[6]. However, conventional techniques, including adhesive bonding and mechanical assembly, are unsuitable for achieving stable connections in transparent materials[7]. Although thermal diffusion bonding and continuous wave laser welding enable glass joining, the high temperatures involved often result in irreversible damage, such as cracks or voids[8,9].

In recent years, femtosecond (fs) laser micro-welding technology has been recognized as an effective approach for joining transparent materials because of its nonlinear absorption effects and localized phase changes induced by ultrahigh peak power[10–14]. In 2005, Tamaki et al.[15] successfully welded two pieces of silica glass using an fs laser, demonstrating the feasibility of ultrafast laser technology for connecting transparent materials. Isamu et al.[16] observed that a high pulse repetition frequency significantly enhanced nonlinear absorption effects in borosilicate glass, leading to elevated temperatures in the laser-irradiated region and increasing the molten phase material, thereby improving the welding efficiency. Huang et al.[17] proposed that multiple scanning cycles resulted in welds without gaps or cracks, enabling complete sealing of transparent materials. Zhang et al.[18] applied ultrafast laser micro-welding to join dissimilar materials and reported that ultra-high pulse energy was generated through holes in the glass, allowing molten metal to flow into these holes and form a mechanical pin structure for secure connection. Despite achieving high-quality connections, optical contact between materials remains a prerequisite, and fixtures are commonly employed[19–22]. However, designing and fabricating a universal fixture is costly and inefficient for practical applications because of the complexity of the process.

In previous research, although we achieved bonding of non-optically contacted glass samples without a fixture, high pulse energy is required due to the low absorption efficiency, and cracks may be generated at the glass interfaces [23]. Thus, we developed a silver ion solution-assisted fs laser welding process to improve energy utilization efficiency, suppress

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the formation of welding cracks, and increase the suitability of liquid-layer-assisted fs laser welding. The mechanical properties of the welds were assessed, and the formation of silver nanoparticles in steam was analyzed to investigate the welding mechanism. Temperature shock and sealability tests were conducted to evaluate the robustness of the welded samples. Additionally, heterogeneous materials were successfully welded, demonstrating potential applications in solar cell packaging. This approach provides an economical and efficient method for achieving high-quality connections of transparent and hard materials in the field of micronano device fabrication.

2. Experimental section

2.1. Sample pretreatment

Commercial silica glass ($20 \times 20 \times 1$ mm^3), sapphire ($20 \times 20 \times 1$ mm^3), and single-crystal silicon ($10 \times 10 \times 0.33$ mm^3) with surface roughness values of 1.41 μm, 1.14 μm, and 1.15 μm respectively were utilized in the laser welding experiments. The samples underwent ultrasonic cleaning in an ethanol solution for 3 min and were subsequently dried in a nitrogen environment. Following this, plasma cleaning was performed using an argon-based device (Ar, v = 1 mL/min, P = 30 W) to render the surface hydrophilic.

2.2. Preparation of the metal ion precursor solution

A 0.08 M silver nitrate aqueous solution was mixed with a 0.06 M sodium citrate aqueous solution at room temperature. Ammonia solution (25 %–28 %) was then gradually added until a clear solution was obtained. In the experiment, 0.5 μL of silver ion mixture was delivered into the sample gap using a pipette.

2.3. Fs laser welding system

A fs laser system (Light Conversion, Pharos PH2–20 W) with an M^2 factor < 1.2, pulse energy stability < 0.5 %, a wavelength of 1030 nm, a pulse duration of 300 fs, and a repetition frequency of 200 kHz was employed for micro-welding. The laser beam, with a diameter of 4 mm, was focused onto the sample surface using a plano-convex lens with a focal length of 100 mm, the focused spot size was about 33 μm. The samples were positioned on a three-dimensional displacement stage, as illustrated in Fig. 1.

2.4. Characterization

The mechanical properties of the welded samples were evaluated using a universal testing machine (ZGDR, DR-509A) to measure shear resistance. Optical microscopy images of the weld seams were obtained using a metallurgical microscope (Leica, DMC4500). The morphology and chemical composition of the welded region were analyzed through scanning electron microscopy (SEM, TESCAN MIRA LMS) and energy-dispersive X-ray spectroscopy (EDS). Thermal shock testing was conducted in a programmable chamber (GWS, TSG-101S-W) within a temperature range of –65 °C to 150 °C, with heating and cooling rates set at 21.5 °C/min. The shock temperature was maintained for 30 min.

3. Results and discussion

3.1. Fs laser welding of non-optical contact silica glasses

Fs laser parameters (pulse energy, scan speed, repetitive scanning passes) significantly influence the mechanical properties of welded samples. Therefore, this section aims to quantify the relationships between these parameters and mechanical performance and identify the optimal parameter combinations for silver ion solution assisted fs laser welding processes.

The fs laser welding of silica glass assisted by a silver ion solution is depicted in Fig. 2a. The presence of the liquid layer effectively minimized the gap between the samples, eliminating the need for a fixture [23]. The sample was scanned using a fs laser to form two weld seams, with a spacing of 200 μm. Laser irradiation of a silver ion solution results in the generation of a substantial number of silver nanoparticles due to multiphoton reduction[24,25]. These nanoparticles were subsequently excited by the fs laser, which induced localized surface plasmon resonance (LSPR) and enhanced the surface electric field. Notably, at the geometric discontinuities of the nanostructures, "hotspots" emerged,

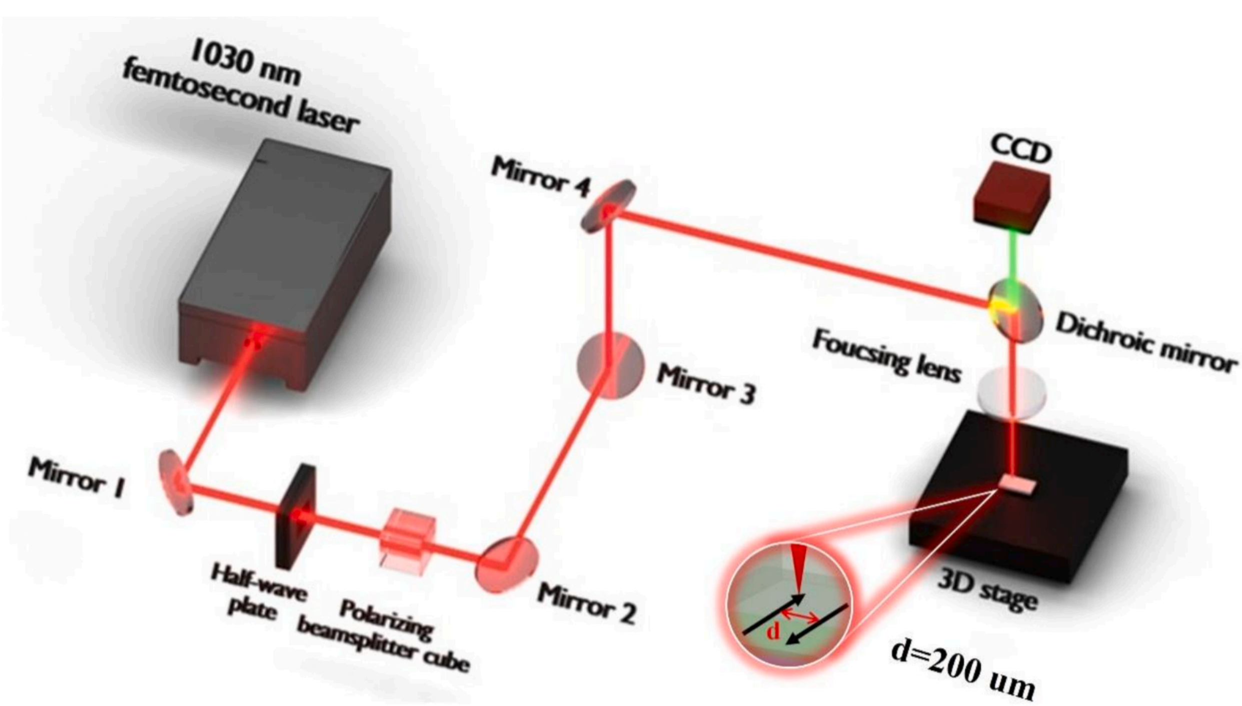

Fig. 1. Schematic representation of the femtosecond laser welding system.

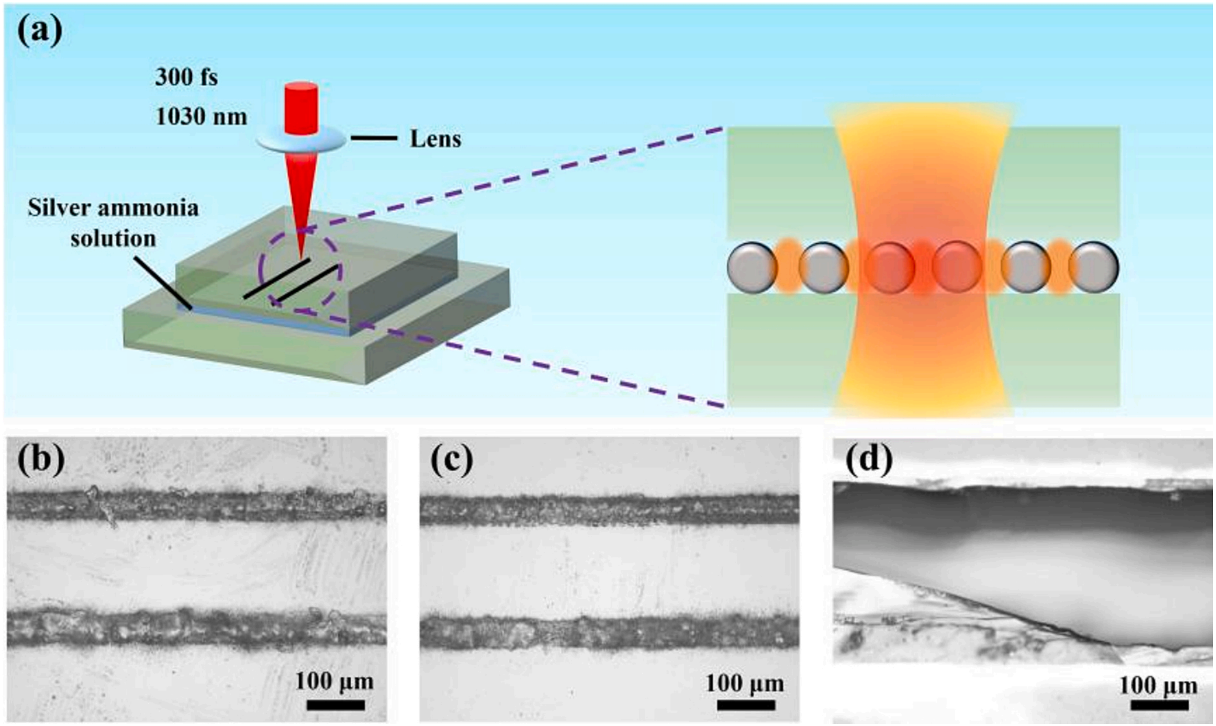

Fig. 2. (a) Schematic of the silver ion solution layer-assisted femtosecond laser micro-welding process; optical images of seams in (b) naturally stacked glasses; (c) a water intermediate layer; and (d) silver ion solution layer-assisted fs laser welding.

leading to localized heating, softening, and melting of the glass[26]. To evaluate welding performance, weld seams were compared under three conditions: natural stacking (Fig. 2b), a water intermediate layer (Fig. 2c), and a silver ion solution (Fig. 2d) at a pulse energy of 40 μJ (The welding processes and performances were shown in Video S1-S3). Insufficient molten glass material under laser-scanned region failed to fill the gap between the seams in the natural stacking conditions, resulting in an unsuccessful glass connection. In Fig. 2c, although the presence of the water layer reduced the gap, effective bonding was not achieved due to low pulse energy. As shown in contrast, as shown in Fig. 2d, a significant amount of molten phase glass material was generated in the laser-scanned region, completely filling the gap between the weld seams and providing strong welding strength. Brittle glass fractures were observed during shear testing, confirming that high-quality bonding of non-optical contact silica glass was successfully achieved through fs laser welding assisted by a silver ion solution.

The effects of pulse energy, scanning speed, and repeated scanning cycles on the mechanical properties of glass welded joints were analyzed. As shown in Fig. 3, the pulse energy has a significant effect on the weld morphology and interface bonding. When the pulse energy is set to 20 μJ (Fig. 3a and e), debris is observed in the laser-scanned region without the formation of a molten glass phase, leading to an ineffective joint. Increasing the pulse energy to 30 μJ (Fig. 3b and f) results in localized melting of the glass, forming a continuous welding area. Brittle fractures are observed after shear testing, and the bonding strength remains relatively low at 4.03 MPa. When the pulse energy is further increased to 40 μJ (Fig. 3c and g), the molten glass fully fills the gap between the seams, forming a dense and continuous weld area and thereby increasing the bonding strength to 27.36 MPa. The shear strength achieved surpasses that reported in previous studies (Table 1), while the input pulse energy density during laser scanning remains significantly lower, indicating that this approach is energy-efficient.

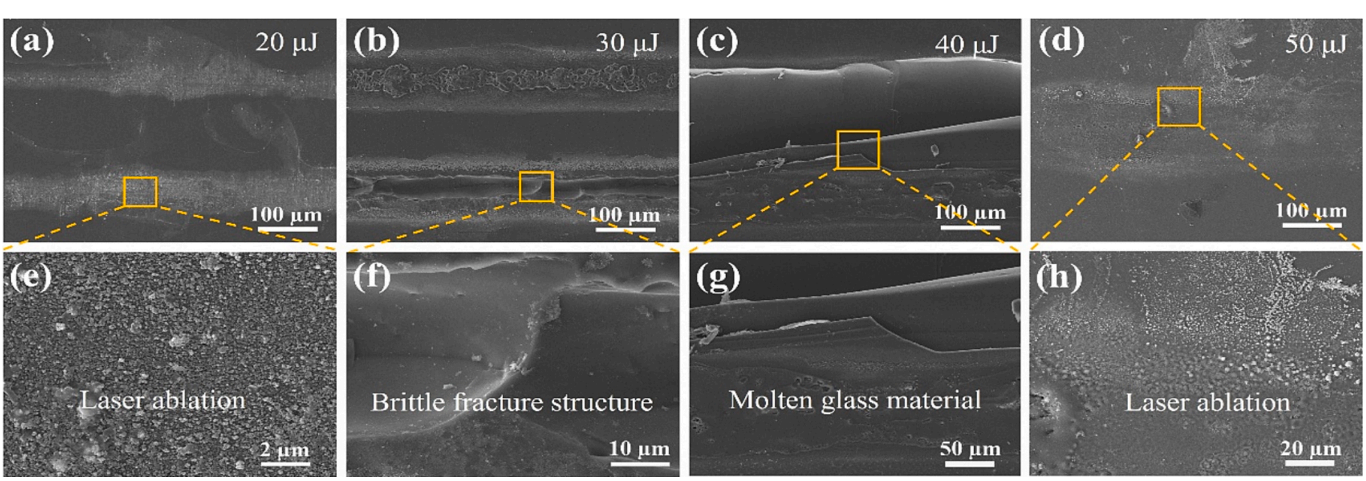

Fig. 3. SEM images of welded joints with different pulse energies. (a, e) 20 μJ; (b, f) 30 μJ; (c, g) 40 μJ; (d, h) 50 μJ.

Table 1
Comparison of shear strength in glass welding using ultrafast laser.

Material	Energy density	Repetition rate	Shear strength	Reference
soda-lime glass	9.42 J/cm^2	50 kHz	6.5 MPa	[19]
aluminosilicate glass	234 J/cm^2	1 MHz	17.4 MPa	[11]
silica glass	2.2 J/cm^2	1 MHz	2 kN	[30]
silica glass	0.86 J/cm^2	500 kHz	1.4 MPa	[31]
silica glass	2.4 J/cm^2	200 kHz	27.36 MPa	This study

However, when the pulse energy exceeds 50 μJ (Fig. 3d and h), noticeable ablation rather than effective melting occurs in the laser-scanned area. The high-energy laser passing through the glass causes the actual focal point to deviate from the expected position on the basis of the linear optical design owing to the self-focusing effect. This focal shift alters the energy density at the interface between the two silica glass pieces during laser welding, leading to the observed ablation phenomenon. Additionally, the temperature gradient induced by the laser may result in surface reconstruction of the material, modifying the light transmission path within the glass[27–29].

The effects of the scanning speed on the welding quality and mechanical properties were further investigated. As shown in Fig. 4, the scanning speed plays a crucial role in determining the dynamic behaviour of the melt pool and the quality of glass bonding. The micromorphological images (Fig. 4a-d) indicate that when the scanning speed exceeds 50 mm/s, the molten glass phase is insufficient to sustain a stable melt pool due to the low heat input. As the scanning speed decreases from 50 mm/s to 1 mm/s, the amount of molten phase increases, allowing the gap between the seams to be completely filled, resulting in a continuous and gap-free weld structure. The mechanical performance testing results (Fig. 4e) demonstrate that the shear strength increases significantly as the scanning speed decreases, rising from 3.15 MPa at 50 mm/s to 27.36 MPa at 1 mm/s. At lower scanning speeds, enhanced thermal accumulation during the welding process strengthens glass bonding. However, although further reductions in the scanning speed may improve the bonding strength, excessive thermal accumulation can lead to defects such as microcracks, reducing the overall efficiency of the process for practical applications.

Fig. 5 illustrates the relationship between the number of repeated scans and the amount of molten glass generated. When a single scan was applied (Fig. 5a and e), insufficient thermal accumulation resulted in the absence of a noticeable molten glass phase, with severe ablation occurring instead. With two scans (Fig. 5b and f), discrete molten phases begin to form in localized regions, and the amount of molten glass tend to increase trend as the number of scans increases (Fig. 5c and g). Upon reaching six scans (Fig. 5d and h), a continuous and dense weld structure was established. Mechanical testing indicated that as the number of repeated scans increased from 1 to 24, the shear strength improved from 3.12 MPa to 27.36 MPa, representing a 780 % increase (Fig. 3i). This increase in bonding strength can be attributed to the thermal accumulation effect generated by multiple scans, which facilitates the formation and accumulation of molten glass material. Additionally, the repeated thermal cycles induced by laser scanning aid in relieving residual stress within the weld passes and suppress the formation of microcracks, further improving the integrity of the welded structure. Considering the welding efficiency, we did not perform the welding process with more than 24 scan repetitions.

3.2. Mechanism of silver ion solution layer-assisted fs laser welding

In the fs laser welding process, the photoionization of water molecules in the precursor solution and the multiphoton reduction of metal ions play crucial role in enhancing the welding quality. As illustrated in Fig. 6a, when the fs laser is focused into an aqueous solution, the laser pulse induces multiphoton ionization of water, generating a dense localized plasma in the solution. This process produces a large number of free electrons and high-energy free radicals, including hydrogen radicals (·H) and hydroxyl radicals (·OH)[32,33]. These free electrons and radicals are subsequently captured by silver ions present in the solution, leading to the formation of silver nanoparticles through the following reaction mechanism:[33–35]

$$Ag^{+} + H_2O \xrightarrow{h\nu} Ag^{0} + H^{+} + \cdot OH$$

$$nAg^{0} \rightarrow (Ag^{0})_m$$

However, the presence of hydroxyl radicals (·OH) in the solution can spontaneously generate H_2O_2, leading to the reoxidation of silver atoms into silver ions, thereby hindering the continuous formation of silver nanoparticles. To counteract this effect, sodium citrate was introduced to eliminate ·OH radicals. Additionally, sodium citrate also acts as a reducing agent in the reaction, facilitating the reduction process, as illustrated in Fig. 6b):[36]

$$C_6O_7H_5^{3-} + 2Ag(NH_3)_2^{+} + OH^{-} \xrightarrow{2h\nu} C_5O_5H_4^{2-} + CO_2 + H_2O + 2Ag$$

Additionally, when fs laser energy is deposited onto the glass surface, a plasma plume is generated, producing various reactive species,

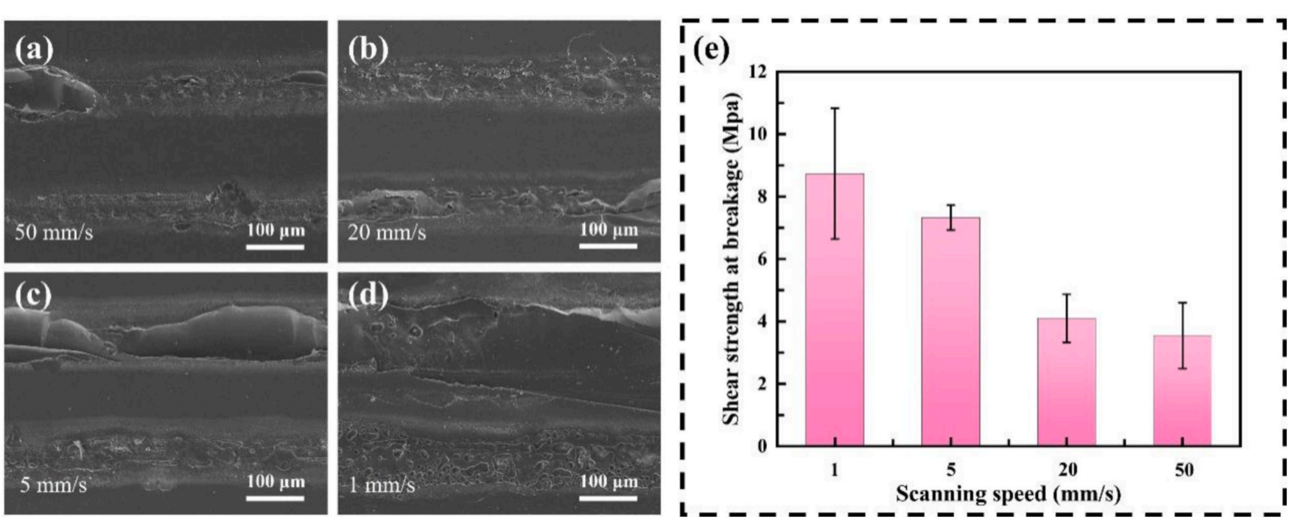

Fig. 4. Effects of the scanning speed on the microstructures of the welded joints and their mechanical properties. SEM images of joints welded at (a) 50 mm/s; (b) 20 mm/s; (c) 5 mm/s; (d) 1 mm/s; (e) shear strength of welded samples at different scanning speeds.

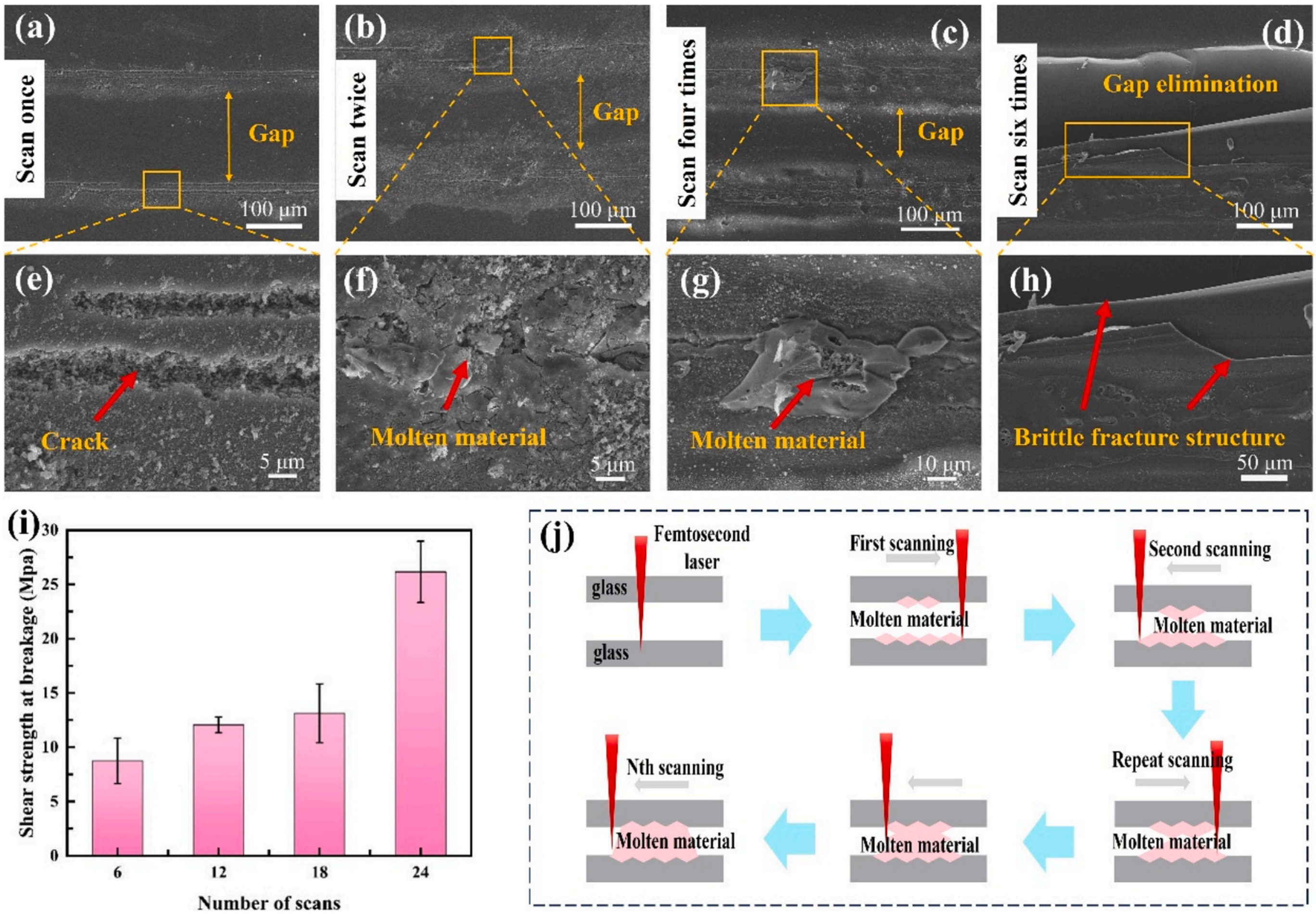

Fig. 5. Effects of repeated scanning cycles on the microstructures of welded joints and their mechanical properties. (a, e) SEM images of the welded joint after 1 scanning repetition; (b, f) 2 repetitions; (c, g) 4 repetitions; (d, h) 6 repetitions; (i) shear strength of welded samples under different scanning repetitions; (j) schematic illustration of the welding mechanism under multiple scanning repetitions.

including electrons, ions, atoms, and clusters. These reactive species rapidly disperse into the solvent, facilitating the reduction of silver ions to atoms at the plasma/liquid interface. This process promotes the continuous formation of silver nanoparticles, as illustrated in Fig. 6c [37].

As shown in Fig. 7a, the welded joint exhibited a high-density distribution of granular nanoparticles that were encapsulated within the molten glass during the laser scanning process. EDS analysis confirmed that these nanoparticles primarily consisted of elemental silver (Ag), providing evidence for the in situ formation of Ag nanoparticles during laser irradiation. Notably, significant deposition of silver nanoparticles also occurs on both sides of the welded zone. These nanoparticles form densely packed, thin silver layers on the glass surface. The thermal conductivity at the metal-amorphous material interface has been demonstrated to be significantly enhanced through the combined effects of elastic and inelastic phonon scattering[38].

The ultraviolet–visible-near infrared (UV–Vis-NIR) absorption spectra indicate that the welded glasses exhibit a distinct absorption peak at 398 nm, which is absent in intrinsic glass (Fig. S1a). This peak is strongly associated with the dipole oscillation mode of the silver nanoparticles, suggesting a resonant interaction between the incident laser radiation and the collective oscillations of surface-free electrons in the Ag nanoparticles. The interactions significantly enhance the optical absorption efficiency within specific spectral bands, particularly in the near-ultraviolet region. Additionally, secondary absorption peaks appear at 553 nm and 687 nm, attributed to the excitation of higher-order plasmonic resonance modes induced by fs laser-driven morphological modifications of Ag nanoparticles. These peaks also result from near-field enhancement effects caused by interparticle spacing below critical coupling distances [39,40]. The sample exhibited characteristic absorption peaks at 398 nm, 553 nm, and 687 nm, where silver nanoparticles significantly enhanced the local electric field with enhancement factors of 78.6, 12, and 9.79, respectively. Despite using a 1030 nm fs laser source, its two-photon absorption process enables efficient excitation of the 553 nm absorption peak, whereas the three-photon absorption process achieves effective matching with the 398 nm absorption peak through energies relaxation (Fig. S1b-c). Consequently, the nanoparticles efficiently generate transient high temperatures through the photothermal effect, providing sufficient energy to facilitate silica glass softening at relatively low fs laser pulse energy.

As shown in Figs. S2a-c, the size of individual silver nanoparticles increases from 46 nm at an energy density of 1.02J/cm^2 to 76 nm at an energy density of 2.4J/cm^2. Moreover, the localized enhanced electric field increases by a factor of 3.22, as illustrated in Figs. S2d-f, indicating that silver nanoparticles facilitate the melting of glass materials, thereby improving the bonding strength. Fig. S2g-i shows the localized electric field enhancement induced by different arrangements of silver nanoparticles. The magnitude of the enhancement increases with increasing number of nanoparticles. Compared with the square arrangement, the hexagonal configuration generates a uniform electric field distribution, facilitating localized softening and melting of glass materials. To further analyze the influence of silver nanoparticles on the temperature distribution in fs laser welding, a transient thermal model of fs laser welding for glass substrates was employed to simulate the temperature field in

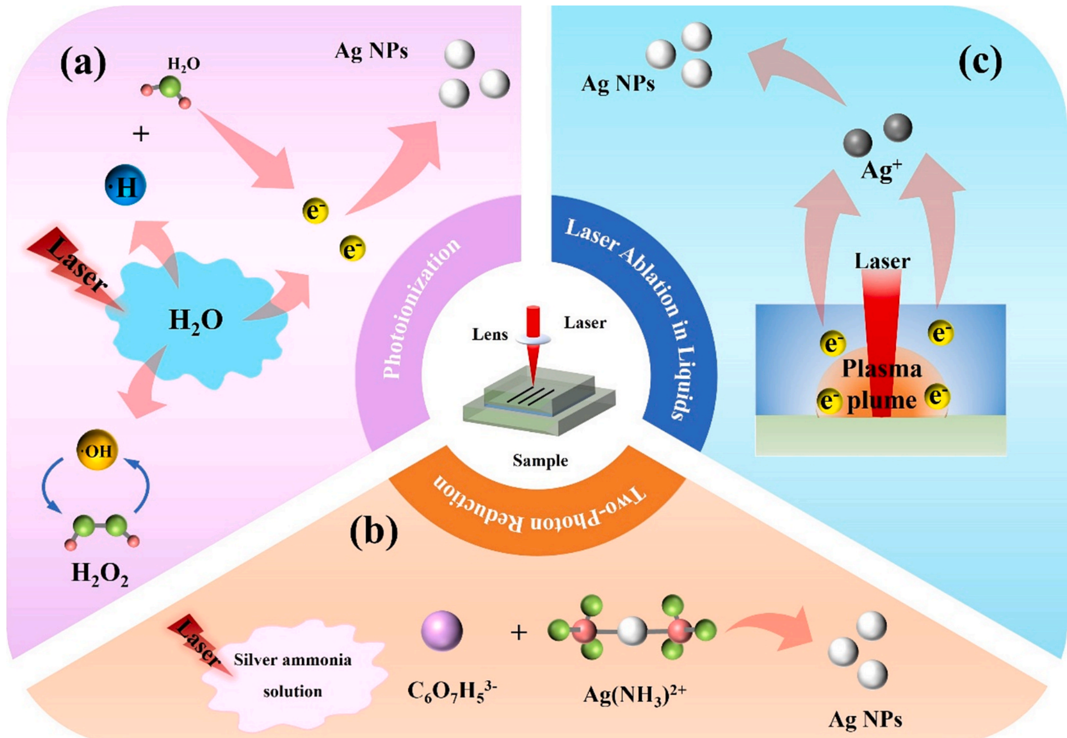

Fig. 6. Mechanisms of fs laser reduction of silver ions to form nanoparticles. (a) Fs laser-induced multiphoton ionization of water; (b) Fs laser irradiation induces multiphoton reduction; (c) Laser ablation in liquid.

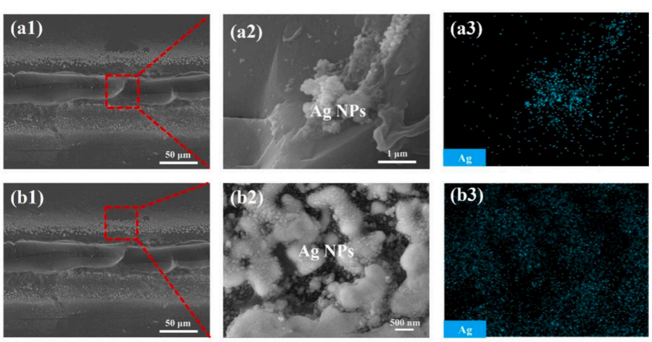

Fig. 7. (a) Micromorphology and elemental characterization of the welded area; (b) micromorphology and elemental characterization of both sides of the welded area.

both glass-glass and glass-silver-glass samples (Fig. 8a and b). The laser beam used in the experiment follows a Gaussian spatial distribution, and the laser intensity at the incident surface (z = 0) is expressed as follows: [41]

$$I(x,y,t) = I_0 exp\left(-\frac{2R^2}{r^2}\right)$$

where, I_0 represents the peak light intensity, R denotes the position of the beam spot, given by $R = \sqrt{(x-vt)^2 + y^2}$, and r represents the radius of the beam spot. For simplicity, the thermal properties of the glass, including density, thermal conductivity, and specific heat at constant pressure, are assumed to be temperature-independent. Fig. 8c shows the temperature variation over time at the centre of the laser scanning area. After 0.5 s (corresponding to a laser spot movement of 0.5 mm), the sample containing the silver layer reaches a peak temperature of 2130 K, which exceeds the melting point of silica glass (about 2000 K). In contrast, the maximum temperature in the glass-glass model is only 1760 K. These results indicate that silver nanoparticles play a significant role in facilitating heating, melting, and solidification processes, ensuring better control over both molten pool formation and solidification behaviour.

3.3. Fs laser micro-welding in solar cell chip packaging

Since welded samples may be subjected to harsh environments in practical applications, thermal shock and waterproof tests were conducted separately. Fig. 9a shows the variation in the shear strength of the silica glass welded samples under high-low-temperature shock. The t shear strength gradually decreases with increasing shock duration. Specifically, after 24 h of high-low-temperature shock, the shear strength of the welded samples is 7.69 MPa, which is a minimal change compared with that of the initial welded samples (8.73 MPa). This finding indicates that silica glass welded with a silver ion solution retains its mechanical properties and resistance to thermal shock over a one-day period. However, with prolonged shock exposure, the shear strength decreases further, reaching 2.56 MPa after 96 h. The reduction in shear strength is attributed to the mismatch in the coefficient of thermal expansion at the silica glass welded joint interface, which induces the accumulation of thermal stress and the formation of microcracks during temperature fluctuations. These microcracks become more pronounced due to repeated thermal shock, ultimately causing a substantial decline in the shear strength of the welded joint.

For the sealability test, the welded samples were immersed in a beaker containing red ink, as shown in Fig. 9b. After two weeks, no penetration of the red liquid into the sealed region was observed (inset in Fig. 9b), confirming the excellent sealing performance. The welded samples were subsequently submerged at a depth of 1 m underwater for 30 min (Fig. 9c-e). Despite the application of high water pressure, no red dye infiltrated the welded area. These results demonstrate that the welded samples maintain effective sealing not only under normal conditions but also in underwater environments, meeting the IPX7

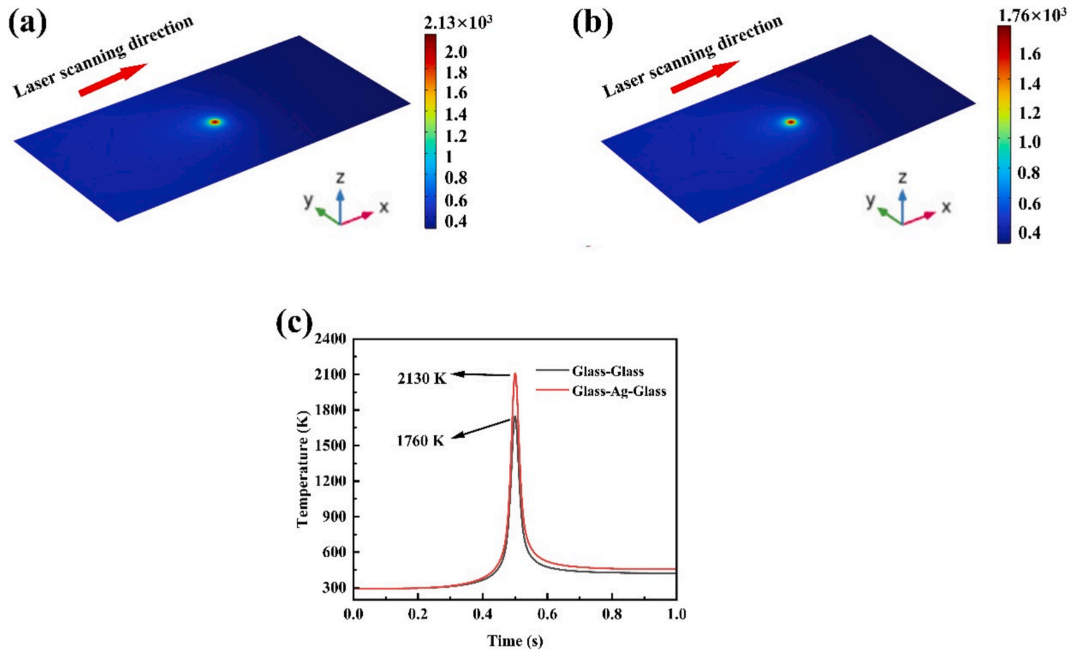

Fig. 8. Temperature distribution characteristics along the weld centreline during the laser scanning process. (a) Glass-silver-glass; (b) glass-glass; (c) temperature variation over time at the central point of the laser scanning region.

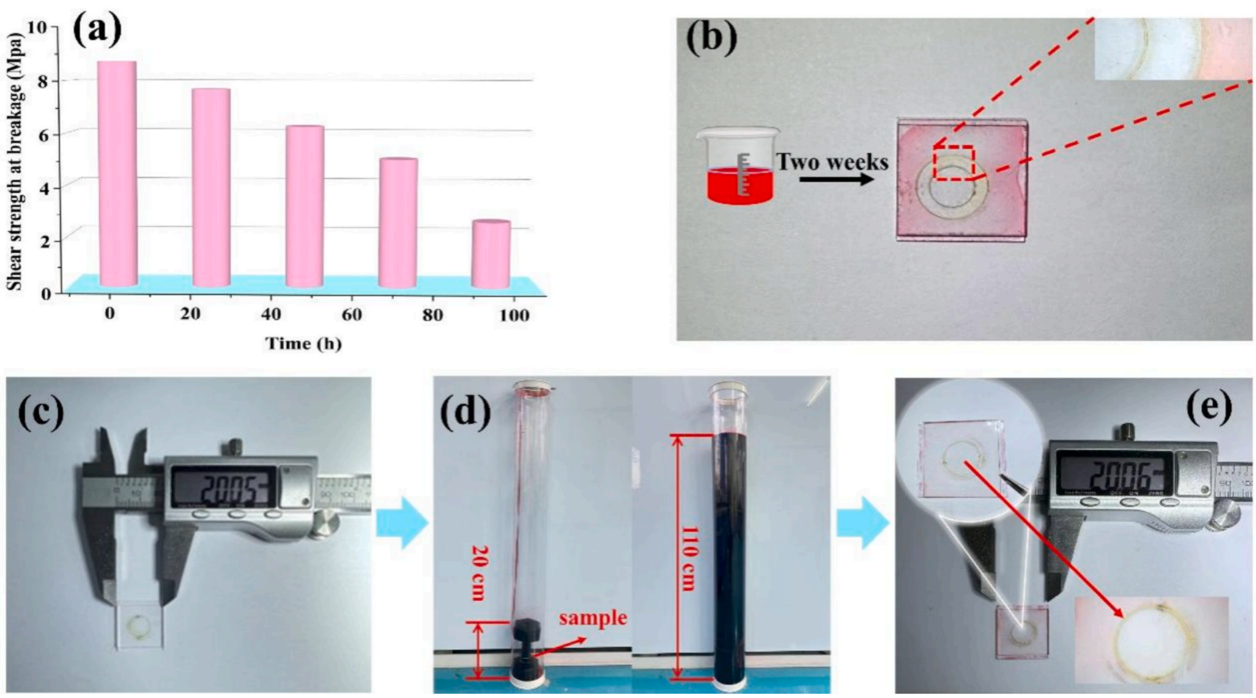

Fig. 9. (a) Variation in shear strength under different durations of temperature shock; (b) sealing test; (c) Welded sample before the negative pressure sealability test; (d) Diving tank used for the negative pressure sealability test; (e) Welded sample after the negative pressure sealability test.

waterproof standard (IEC 60529:2013, IDT) for enclosure packaging in practical applications.

In recent years, fs laser welding of heterogeneous materials has gradually become a research hotspot[42], and our proposed method was also demonstrated to be effective in joining heterogeneous materials. Monocrystalline silicon and sapphire were selected as representative semiconductor and optical materials due to their distinct thermophysical properties. Despite the material property mismatches, a successful heterojunction was achieved through fs laser welding, as shown in Fig. 10a and b. Furthermore, hermetic encapsulation of photovoltaic cells was performed using welded glass (Fig. 10c and d). A photovoltaic cell contacted by conductive adhesive tape is positioned within the recess of a quartz glass substrate for encapsulation. To enable electrical signal monitoring, the top interface of the sealed structure remains intentionally unwelded. The packaged solar cell chips retained their conductivity in water, demonstrating their ability to operate underwater (Fig. 10e and f). These findings indicate that the silver ion solution interlayer-assisted fs laser welding process enables the formation of

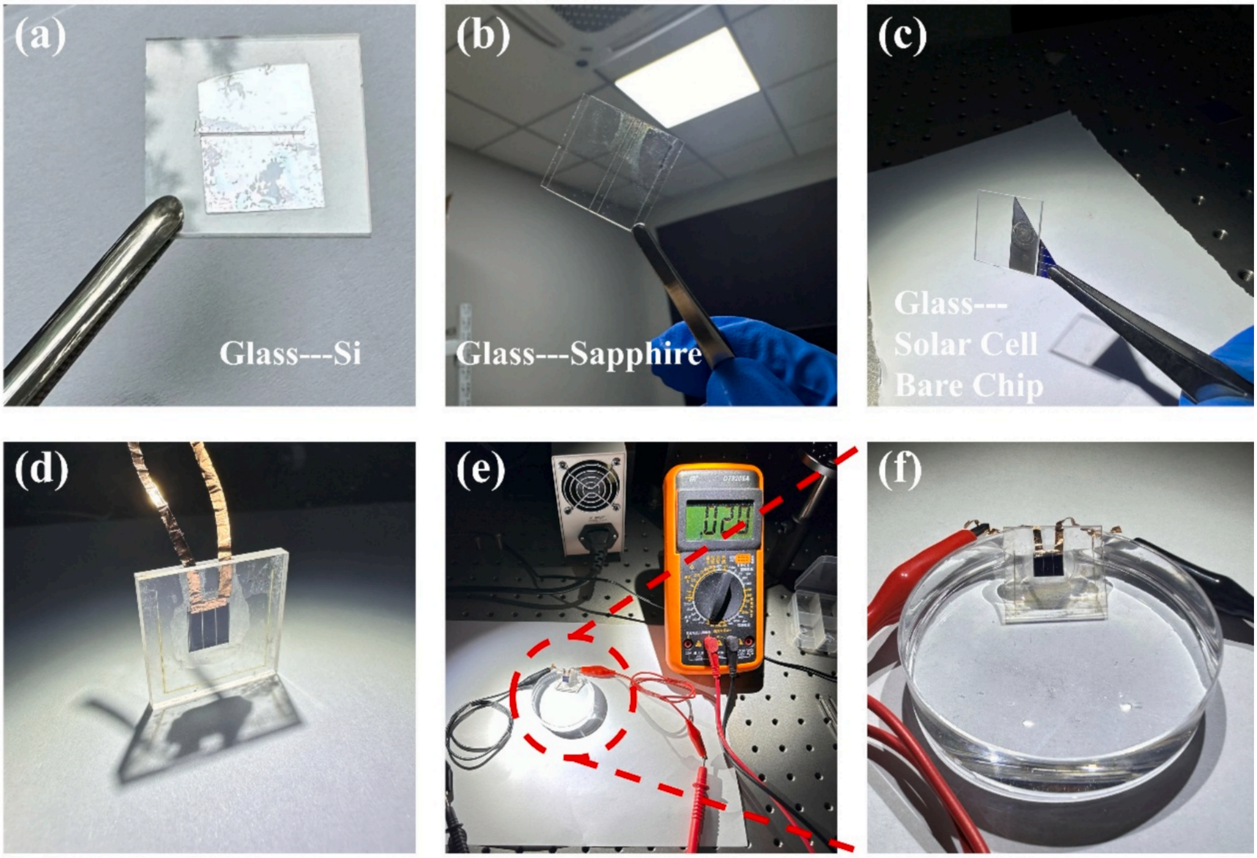

Fig. 10. Photographic images of the (a) glass-silicon connection; (b) glass-sapphire connection; (c) glass-solar-cell bare-chip connection; (d) encapsulated solar-cell bare-chip device; (e, f) conductivity test of the sealed device immersed in water.

high-strength connections while effectively mitigating the impact of moisture and other extreme environmental factors on the performance of solar devices.

4. Conclusion

This study introduced a novel approach for fs laser welding of transparent and hard materials using a silver ion solution, which facilitates interface modification through fs laser-induced reduction of silver nanoparticles. The findings demonstrated that localized optical field enhancement, driven by plasmon resonance and interface light absorption mechanisms, significantly increased the photon absorption efficiency of the fs laser energy. Microstructural characterization confirmed the formation of a uniform and continuous connecting seam through fs laser scanning. Mechanical testing indicated that the shear strength of the welded joint reached 27.36 MPa, with the potential for further improvement through multiple scanning cycles. The reliability of this method was validated through thermal shock and water sealability tests, which achieved compliance with the IPX7 waterproof standard (IEC 60529:2013, IDT). This silver ion solution-assisted laser welding strategy offers an effective solution for overcoming the challenges of limited photon energy absorption efficiency and bonding non-optically contacted transparent and hard materials.

CRediT authorship contribution statement

Hao Chen: Writing – original draft, Investigation, Formal analysis, Data curation. **Zhaoxu Li:** Formal analysis. **Mingyang Han:** Data curation. **Xiao Yang:** Funding acquisition. **Zhimin Liang:** Supervision. **Shi Bai:** Writing – review & editing, Supervision, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.matdes.2025.114356. Fig. S1 shows the UV–Vis-NIR absorption spectra of the glass substrate and welded glasses. Fig. S2 shows the influence of laser power on the size of the reduced silver nanoparticles during the welding process, along with the degree of localized electric field enhancement achieved by silver nanoparticles with varying dimensions. Video S1 shows the fs laser welding of natural stacked glasses. Video S2 shows the water assisted fs laser welding of glasses. Video S3 shows silver ion solution assisted fs laser welding of glasses.

Data availability

The datasets generated and analyzed during the current study are available from the corresponding author upon reasonable request.

References

[1] J. Zhang, S. Chen, H. Lu, M. Huang, J. Li, L. Guo, Q. Lue, Q. Zhang, The effect of gap on the quality of glass-to-glass welding using a picosecond laser, Opt. Lasers Eng. 134 (2020) 106248.

[2] S. Richter, F. Zimmermann, A. Tünnermann, S. Nolte, Laser welding of glasses at high repetition rates–Fundamentals and prospects, Opt. Laser Technol. 83 (2016) 59–66.

[3] I. Miyamoto, K. Cvecek, M. Schmidt, Advances of Laser Welding Technology of Glass-Science and Technology, J. Laser Micro/Nanoeng. 15 (2) (2020) 63–76.

[4] H. Chen, Y. Rong, Y. Huang, C. Wu, Crack suppression of glass welding by ultrafast laser without optical contact based on light modulation, Opt. Laser Technol. 164 (2023) 109466.

[5] L. Schaefer, M. Schmidt, Welding of glass fibres onto large-scale substrates with high mechanical stability and optical quality, Phys. Procedia 5 (2010) 145–152.

[6] D. Lei, Z. Wang, J. Li, The calculation and analysis of glass-to-metal sealing stress in solar absorber tube, Renew. Energy 35 (2) (2010) 405–411.

[7] C. Li, J. Tan, M. Luo, W. Chen, Y. Huang, J. Gu, N. Zhao, J. Li, H. Yang, Q. Zhang, High shear strength welding of soda lime glass to stainless steel using an infrared nanosecond fiber laser assisted by surface tension, Opt. Lasers Eng. 161 (2023) 107329.

[8] C. Tian, H. Ren, H. Shen, The connection of glass and metal with a large gap by combining laser soldering and ultrafast laser welding, J. Manuf. Process. 102 (2023) 528–534.

[9] D. Nodop, P. Wiemuth, J. Rücker, M. Kahle, Glass Welding with Ultrashort Laser Pulses Producing mm-long Filaments, J. Laser Micro/Nanoeng. 15 (3) (2020).

[10] O.P. Ciuca, R.M. Carter, P.B. Prangnell, D.P. Hand, Characterisation of weld zone reactions in dissimilar glass-to-aluminium pulsed picosecond laser welds, Mater Charact 120 (2016) 53–62.

[11] G. Zhang, Y. Pan, P. Wu, Z. Guo, J. Lv, H. Zhang, J. Wang, W. Zhang, J. Xu, L. Wang, Glass micro welding in thermal accumulation regime with using spatially shaped ultrafast laser, Opt. Laser Technol. 168 (2024) 109845.

[12] S. Richter, S. Döring, A. Tünnermann, S. Nolte, Bonding of glass with femtosecond laser pulses at high repetition rates, Appl. Phys. A 103 (2011) 257–261.

[13] M. Shimizu, M. Sakakura, M. Ohnishi, Y. Shimotsuma, T. Nakaya, K. Miura, K. Hirao, Mechanism of heat-modification inside a glass after irradiation with high-repetition rate femtosecond laser pulses, J. Appl. Phys. 108 (7) (2010) 073533.

[14] J. Xu, Q. Jiang, J. Yang, J. Cui, Y. Zhao, M. Zheng, J. Oliveira, Z. Zeng, R. Pan, S. Chen, A review on ultrafast laser microwelding of transparent materials and transparent material–metals, Metals 13 (5) (2023) 876.

[15] T. Tamaki, W. Watanabe, J. Nishii, K. Itoh, Welding of transparent materials using femtosecond laser pulses, Jpn. J. Appl. Phys. 44 (5L) (2005) L687.

[16] I. Miyamoto, A. Horn, J. Gottmann, D. Wortmann, F. Yoshino, Fusion welding of glass using femtosecond laser pulses with high-repetition rates, J. Laser Micro/Nanoeng. 2 (1) (2007) 57–63.

[17] H. Huang, L.-M. Yang, J. Liu, Direct welding of fused silica with femtosecond fiber laser, SPIE, Laser-Based Micro-and Nanopackaging and Assembly Vi, 2012, pp. 19–27.

[18] L. Zhang, H. Wu, J. Wen, M. Li, X. Shao, X. Ma, Glass to aluminum joining by forming a mechanical pin structure using femtosecond laser, J. Mater. Process. Technol. 302 (2022) 117504.

[19] X. Jia, K. Li, Z. Li, C. Wang, J. Chen, S. Cui, Multi-scan picosecond laser welding of non-optical contact soda lime glass, Opt. Laser Technol. 161 (2023) 109164.

[20] R. Pan, D. Yang, T. Zhou, Y. Feng, Z. Dong, Z. Yan, P. Li, J. Yang, S. Chen, Micro-welding of sapphire and metal by femtosecond laser, Ceram. Int. 49 (13) (2023) 21384–21392.

[21] Y. Feng, R. Pan, T. Zhou, Z. Dong, Z. Yan, Y. Wang, P. Chen, S. Chen, Direct joining of quartz glass and copper by nanosecond laser, Ceram. Int. 49 (22) (2023) 36056–36070.

[22] S. Zhang, R. Pan, W. Wang, X. Zhang, T. Zhou, X. Li, Y. Feng, R. Xu, T. Lin, P. He, Direct welding of dissimilar ceramics YSZ/Sapphire via nanosecond laser pulses, J. Eur. Ceram. Soc. 44 (7) (2024) 4782–4796.

[23] H. Chen, Z. Li, M. Han, X. Yang, S. Bai, Improved Femtosecond Laser Welding of Nonoptical Contact Glass by pressure and a Water Intermediate Layer, ACS Omega 9 (37) (2024) 38878–38886.

[24] B.-B. Xu, Z.-C. Ma, L. Wang, R. Zhang, L.-G. Niu, Z. Yang, Y.-L. Zhang, W.-H. Zheng, B. Zhao, Y. Xu, Localized flexible integration of high-efficiency surface enhanced Raman scattering (SERS) monitors into microfluidic channels, Lab Chip 11 (19) (2011) 3347–3351.

[25] R. Xue-Liang, Z. Mei-Ling, J. Feng, Z. Yuan-Yuan, D. Xian-Zi, L. Jie, Y. Hong, D. Xuan-Ming, Z. Zhen-Sheng, Laser Direct writing of Silver Nanowire with Amino Acids-Assisted Multiphoton Photoreduction, J. Phys. Chem. C 120 (46) (2016) 26532–26538.

[26] X. Wang, J. Cui, J. Zhang, W. Wang, X. Mei, W. Long, H. Sun, X. He, H. Xie, Simulation study of near-field enhancement on an Ag nanoparticle dimer system in a laser-induced nanowelding process, Integr. Ferroelectr. 191 (1) (2018) 72–79.

[27] Z. Yang, C. Tian, H. Ren, X. Wei, H. Shen, Welding threshold in ultrafast laser welding of quartz glass and 304 stainless steel, Opt. Laser Technol. 181 (2025) 111622.

[28] C. Wang, S. Zhang, Z. Luo, K. Ding, B. Liu, J.a. Duan, High-quality welding of glass by a femtosecond laser assisted with silver nanofilm, Applied Optics 60(18) (2021) 5360-5364.

[29] J. Zhan, Y. Gao, J. Sun, W. Zhu, S. Wang, L. Jiang, X. Li, Mechanism and optimization of femtosecond laser welding fused silica and aluminum, Appl. Surf. Sci. 640 (2023) 158327.

[30] M. Kahle, D. Nodop, P. Wiemuth, Glass-welding with USP-lasers and long working distances, SPIE, Eighth European Seminar on Precision Optics Manufacturing, 2021.

[31] S. Kim, J. Kim, Y.-H. Joung, J. Choi, C. Koo, Bonding strength of a glass microfluidic device fabricated by femtosecond laser micromachining and direct welding, Micromachines 9 (12) (2018) 639.

[32] C. Li, J. Hu, L. Jiang, C. Xu, X. Li, Y. Gao, L. Qu, Shaped femtosecond laser induced photoreduction for highly controllable Au nanoparticles based on localized field enhancement and their SERS applications, Nanophotonics 9 (3) (2020) 691–702.

[33] V.K. Meader, M.G. John, L.M. Frias Batista, S. Ahsan, K.M. Tibbetts, Radical chemistry in a femtosecond laser plasma: photochemical reduction of Ag^{+} in liquid ammonia solution, Molecules 23 (3) (2018) 532.

[34] T. Nakamura, H. Magara, Y. Herbani, S. Sato, Fabrication of silver nanoparticles by highly intense laser irradiation of aqueous solution, Appl. Phys. A 104 (2011) 1021–1024.

[35] H. Zeng, X.W. Du, S.C. Singh, S.A. Kulinich, S. Yang, J. He, W. Cai, Nanomaterials via laser ablation/irradiation in liquid: a review, Adv. Funct. Mater. 22 (7) (2012) 1333–1353.

[36] S. Bai, H. Chen, Z. Li, S. Kawabata, Z. Song, K. Sugioka, Ultrafast Laser Nanoprocessing and applications, in: N. Kalarikkal (Ed.), Laser-Based Techniques for Nanomaterials: Processing to Characterization, Royal Society of Chemistry London, 2024, pp. 95–129.

[37] P. Ran, L. Jiang, X. Li, B. Li, P. Zuo, Y. Lu, Femtosecond Photon-Mediated Plasma Enhances Photosynthesis of Plasmonic Nanostructures and their SERS applications, Small 15 (11) (2019) 1804899.

[38] J. El Hajj, C. Adessi, M. de San Feliciano, G. Ledoux, S. Merabia, Enhanced thermal conductance at interfaces between gold and amorphous silicon and between gold and amorphous silica, Phys. Rev. B 110 (11) (2024) 115437.

[39] L.O. Herrmann, V.K. Valev, C. Tserkezis, J.S. Barnard, S. Kasera, O.A. Scherman, J. Aizpurua, J.J. Baumberg, Threading plasmonic nanoparticle strings with light, Nat. Commun. 5 (1) (2014) 4568.

[40] Z. Yin, Y. Wang, C. Song, L. Zheng, N. Ma, X. Liu, S. Li, L. Lin, M. Li, Y. Xu, Hybrid Au–Ag nanostructures for enhanced plasmon-driven catalytic selective hydrogenation through visible light irradiation and surface-enhanced Raman scattering, J. Am. Chem. Soc. 140 (3) (2018) 864–867.

[41] Y. Jia, Y. Saadlaoui, H. Hamdi, J. Sijobert, J.-C. Roux, J.-M. Bergheau, An experimental and numerical case study of thermal and mechanical consequences induced by laser welding process, Case Stud. Therm. Eng. 35 (2022) 102078.

[42] X. Zhang, Y. Zhao, M. Wu, M. Yang, T. Zhang, L. Zhang, J. Yang, C. Tan, X. Song, Ultrafast laser packaging of glass and silicon: Microsturcture and mechanical property, Opt. Laser Technol. 184 (2025) 112538.